

AER1319: FINITE VOLUME METHODS FOR CFD

Assignment #1

For the first assignment, you are required to determine the eigenstructure (i.e., eigenvalues and eigenvectors) of the x -direction and y -direction flux Jacobians, $\mathcal{A}$ and $\mathcal{B}$, and coefficient matrices, $\mathbf{A}$ and $\mathbf{B}$, respectively, for the two-dimensional (2D) form of the Euler equations given by

$$\frac{\partial \mathbf{U}}{\partial t} + \mathcal{A} \frac{\partial \mathbf{U}}{\partial x} + \mathcal{B} \frac{\partial \mathbf{U}}{\partial y} = \mathbf{0}, \quad \frac{\partial \mathbf{W}}{\partial t} + \mathbf{A} \frac{\partial \mathbf{W}}{\partial x} + \mathbf{B} \frac{\partial \mathbf{W}}{\partial y} = \mathbf{0},$$

where vectors of conserved and primitive solution variables are given by

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \frac{p}{\gamma - 1} + \frac{\rho(u^2 + v^2)}{2} \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} \rho \\ u \\ v \\ p \end{bmatrix},$$

and where ρ , u , v , p , and γ are the density, x -direction component of the velocity, y -direction component of the velocity, pressure, and specific heat ratio, respectively. Do this by appropriately modifying the following Maple worksheet for the 1D flow case:

`euler1d-eigensys.mws` (Maple worksheet, download from course web site)

`euler1d-eigensys.pdf` (pdf output, download from course web site)

The worksheet determines the eigenstructure for the one-dimensional (1D) form of Euler equations as covered in the lectures. Note that you can also explore using the Maple eigenvector and eigenvalue commands to determine the eigenstructure directly.

If you do not have access to Maple, Matlab has symbolic manipulation capabilities similar to that of Maple and Octave is an open source version of Matlab that is available under Linux and may also have these capabilities. Mathematica is also an option.

Please submit the Maple worksheet when you turn in your assignment and I will discuss with each of you the results.